

Name: _____

Maths Class: _____

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 12 HSC COURSE

Extension 1 Mathematics

HSC Task 2

March 2010

Time Allowed: 70 minutes

Instructions:

- Write your name and class at the top of this page, and on all your answer sheets.
- Hand in your answers attached to the rear of this question sheet.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated are a guide only and may be varied at the time of marking

(For Markers Use Only)

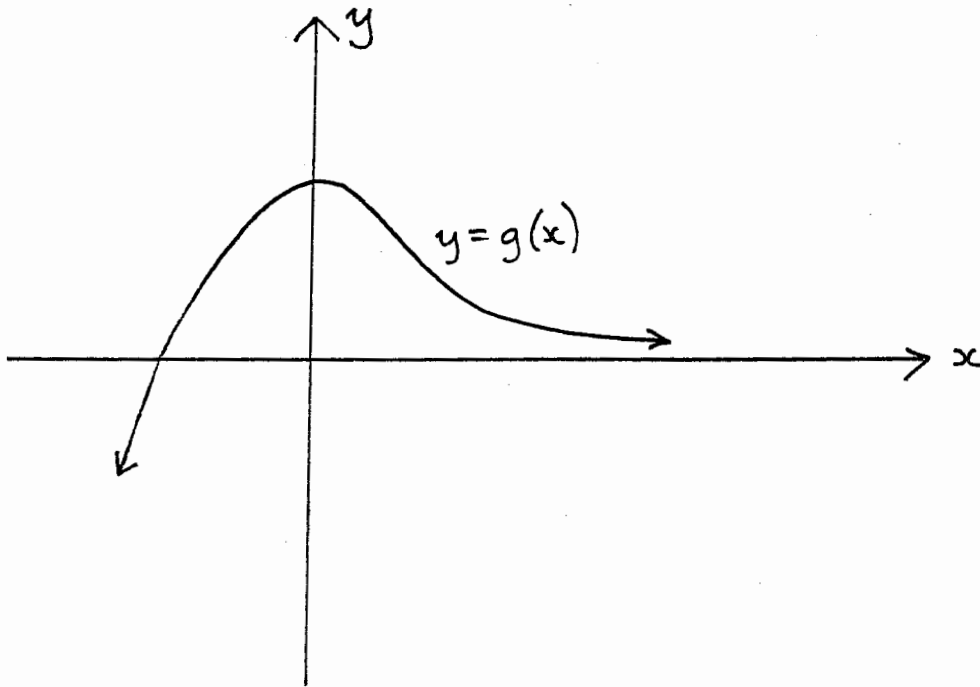
1	2	3	4	5	6	Total
/9	/9	/11	/9	/11	/10	/59

QUESTION 1**(9 marks)**

- a) i) Sketch $y = |1 - 2x|$, for $-1 \leq x \leq 2$ (2)
- ii) Hence, evaluate $\int_{-1}^2 |1 - 2x| dx$ (2)
- b) Sketch a continuous curve $y = f(x)$, in the domain $-4 \leq x \leq 4$, that satisfies all of the following conditions:
- $f(x)$ is odd
 - $f(3) = 0$
 - $f'(1) = 0$
 - $f'(x) > 0$ for $x > 1$
 - $f'(x) < 0$ for $0 \leq x < 1$
- (5)

QUESTION 2**(9 Marks)****(Start a new page)**

- a) The function $y = g(x)$ has been sketched below.



Sketch $y = g'(x)$, the derivative function. (2)

- b) Using a suitable substitution or otherwise find

$$\int x \sqrt{4 - x^2} \, dx \quad (3)$$

c) Show that $\int_0^{\frac{1}{4}} \frac{3x}{(1 + 4x)^3} \, dx = \frac{3}{128}$ (4)

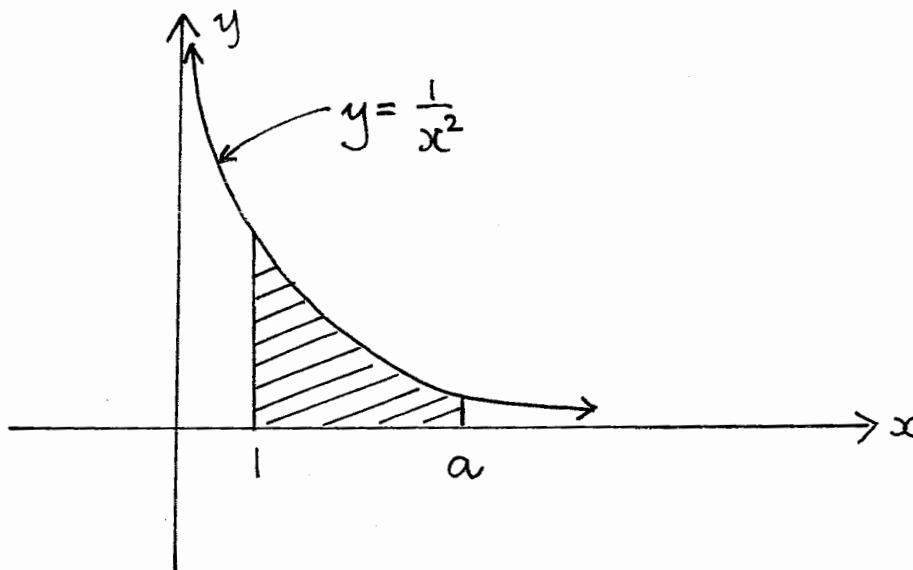
Use the substitution $u = 1 + 4x$

QUESTION 3

(11 Marks) (Start a new page)

- a) The shaded area below is $\frac{2}{3}$ unit². (3)

Find the value of a



- b) i) Show that the function $y = \frac{2x^2}{x^2 + 1}$ has one stationary point and determine its nature. (3)
- ii) Find a horizontal asymptote for this function. (1)
- iii) Sketch the function showing the stationary point and any asymptotes. (label your sketch clearly) (2)
- iv) Without further calculations, indicate with a cross on your sketch, any point(s) of inflexion. (2)

QUESTION 4

(9 Marks)

(Start a new page)

- a) Prove by mathematical induction that

(5)

$3^n + 7^{n+1}$ is divisible by 4 for all positive integers n

- b) An ellipse has the equation $x^2 + 8y^2 = 16$. (The ellipse has its centre at $(0,0)$)

- i) Find where the ellipse cuts the y axis.

(1)

- ii) If the ellipse is rotated around the y axis find the volume of the solid formed. (in exact form)

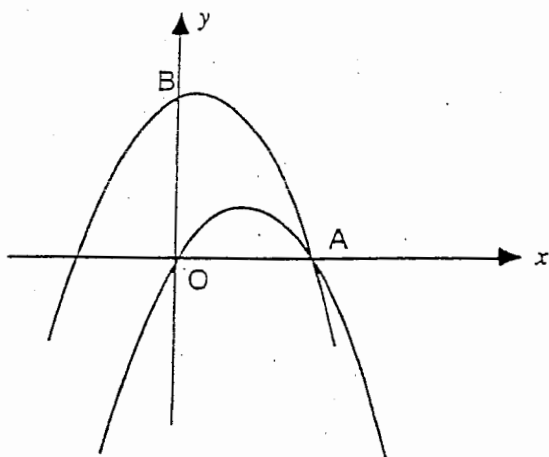
(3)

QUESTION 5

(11 Marks)

(Start a new page)

- a)



The sketch shows the parabolas

$$y = x(3 - x) \text{ and}$$

$$y = (3 - x)(2 + x)$$

- (i) What are the co-ordinates of A and B?

(2)

- (ii) Prove that the area of the region bounded by OB and the arcs OA and AB is equal to that of ΔOAB

(4)

- b) i) Sketch $y = x^2$ and $y = 4x - x^2$ on the same axes and clearly indicate the points intersection.

(2)

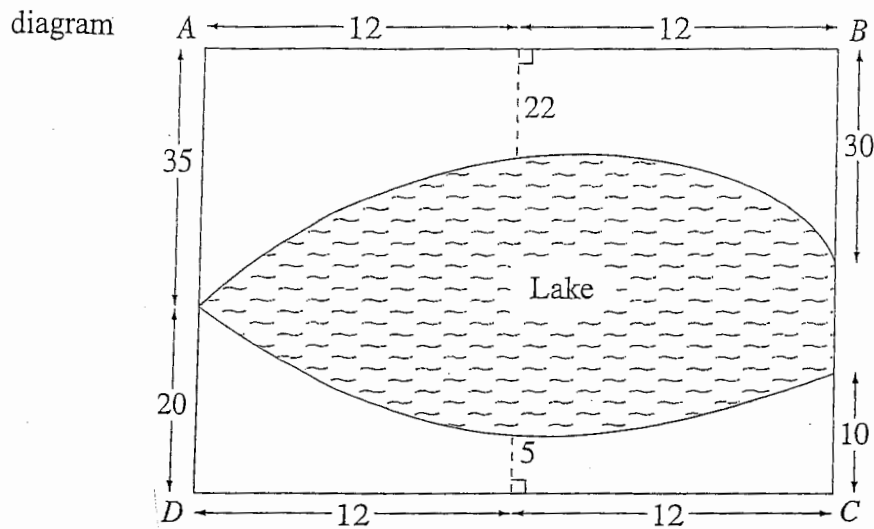
- ii) Find the volume of the hollow cup generated when the area enclosed between the curves $y = x^2$ and $4x - x^2$ is rotated about the x -axis. (in exact form)

(3)

QUESTION 6

(10 Marks) (Start a new page)

- a) There is a lake inside the rectangular grass picnic area $ABCD$, as shown in the

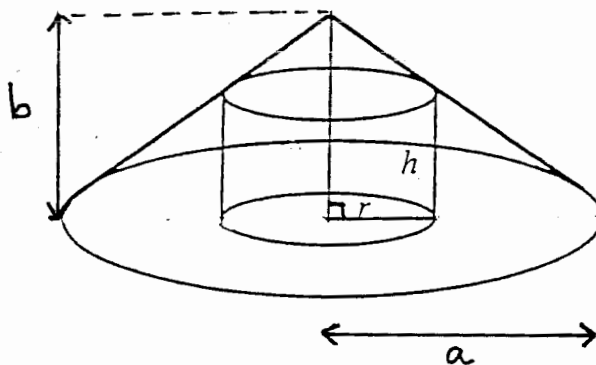


(2)

NOT TO
SCALEAll measurements
are in metres

Use Simpson's Rule to find the approximate area of the lake's surface.

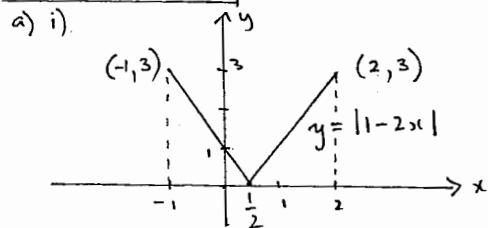
- b)



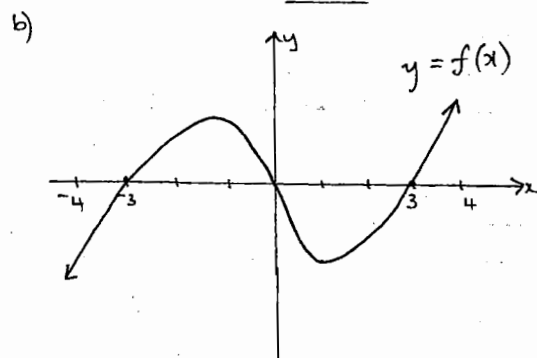
A variable cylinder, radius r and height h , is inscribed in a fixed cone, radius a and height b . (Note: a and b are constants)

- Prove that $h = \frac{b(a-r)}{a}$ (2)
- Express the volume of the cylinder as a function of r (1)
- Find the maximum volume of the cylinder in terms of a and b (4)
- Prove that the cylinder with maximum volume is $\frac{4}{9}$ that of the cone (1)

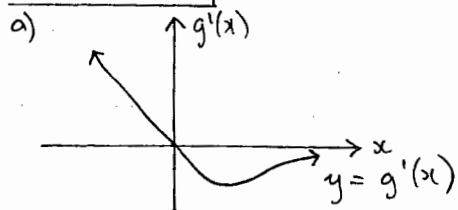
Question 1



ii) $\int_{-1}^2 |1-2x| dx = \frac{2(1\frac{1}{2} \times 3)}{2}$
 $= 4.5$



Question 2



b) $u = 4 - x^2$
 $\frac{du}{dx} = -2x$
 $du = -2x dx$
 $\frac{du}{-2x} = dx$
 $\int x \sqrt{4-x^2} dx = \int x \cdot u^{1/2} \cdot \frac{du}{-2x}$
 $= -\frac{1}{2} \int u^{1/2} du$

$$= -\frac{1}{2} \left[\frac{2u^{3/2}}{3} \right] + C$$

$$= -\frac{1}{3} (4-x^2)^{3/2} + C$$

c) $u = 1+4x \Rightarrow \frac{u-1}{4} = x$
 $\frac{du}{dx} = 4 \therefore \frac{du}{4} = dx$
 $x = \frac{1}{4} \quad u = 2$
 $x = 0 \quad u = 1$
 $\therefore \int_0^{1/4} \frac{3x}{(1+4x)^3} dx = \int_1^2 3 \left(\frac{u-1}{4} \right) \cdot \frac{1}{u^3} \cdot \frac{du}{4}$
 $= \frac{3}{16} \int_1^2 \left(\frac{u-1}{u^3} \right) du$
 $= \frac{3}{16} \int_1^2 (u^{-2} - u^{-3}) du$
 $= \frac{3}{16} \left[-u^{-1} + \frac{u^{-2}}{2} \right]_1^2$
 $= \frac{3}{16} \left[-\frac{1}{u} + \frac{1}{2u^2} \right]_1^2$
 $= \frac{3}{16} \left[\left(-\frac{1}{2} + \frac{1}{8} \right) - \left(-1 + \frac{1}{2} \right) \right]$
 $= \frac{3}{16} \left(-\frac{3}{8} + \frac{1}{2} \right)$
 $= \frac{3}{128}$

Question 3

a) $\int_1^a \frac{1}{x^2} dx = \frac{2}{3}$
 $\int_1^a x^{-2} dx = \frac{2}{3}$
 $\left[-x^{-1} \right]_1^a = \frac{2}{3}$

$$\left[-\frac{1}{x} \right]_1^a = \frac{2}{3}$$

$$-\frac{1}{a} + 1 = \frac{2}{3}$$

$$\frac{1}{3} = \frac{1}{a}$$

$$\therefore a = 3$$

b) i) $u = 2x^2 \quad v = x^2 + 1$
 $u' = 4x \quad v' = 2x$
 $\frac{dy}{dx} = \frac{4x(x^2+1) - 2x \cdot 2x^2}{(x^2+1)^2}$
 $= \frac{4x^3 + 4x - 4x^3}{(x^2+1)^2}$
 $\frac{dy}{dx} = \frac{4x}{(x^2+1)^2}$

st pt $\frac{dy}{dx} = 0 \quad 4x = 0$
 $\therefore x = 0$
 at (0,0) test max/min

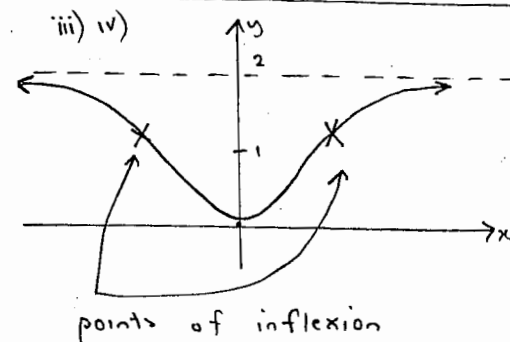
x	-1	0	1
dy/dx	-	0	+

- \quad \min \quad / +
0

$\therefore (0,0)$ is a min turning pt.

ii) $\lim_{x \rightarrow \infty} \frac{2x^2}{x^2+1}$
 $\lim_{x \rightarrow \infty} \frac{x^2(2)}{x^2(1 + \frac{1}{x^2})} = 2$

$\therefore y = 2$ horizontal asymptote

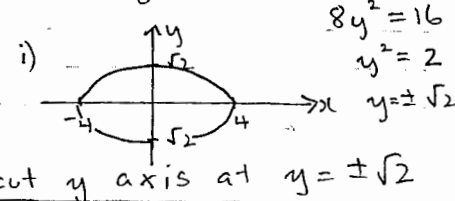


Question 4

" $3^n + 7^{n+1}$ div by 4 positive n"
 Step ① Show true for $n=1$
 $3^1 + 7^2 = 52$ div by 4
 Step ② Assume true for some +ve integer k
 $k \quad k+1$
 $* 3 + 7 = 4M$ (M is an integer)
 Step ③ Prove true for $n=k+1$
 $k+1 \quad k+2 \quad k \quad k^2$
 $3 + 7 = 3 \cdot 3 + 7 \cdot 7$
 (from *)
 $= 3(4M - 7^k \cdot 7) + 49 \cdot 7^k$
 $= 12M - 21 \cdot 7^k + 49 \cdot 7^k$
 $= 12M + 28 \cdot 7^k$
 $= 4(3M + 7 \cdot 7^k)$

Step ④ Since shown true for $n=1$ and if assumed true for $n=k$ we have shown true for $n=k+1 \therefore$ true for all positive integers.

b) $x^2 + 8y^2 = 16$



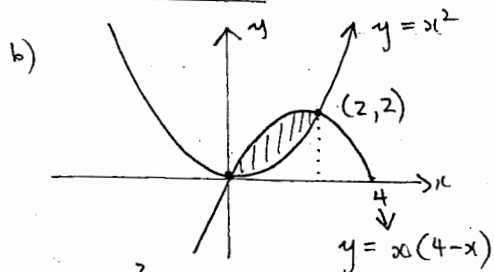
3

$$\begin{aligned}
 V_y &= 2\pi \int_0^{\sqrt{2}} (16 - 8y^2) dy \\
 &= 2\pi \left[16y - \frac{8y^3}{3} \right]_0^{\sqrt{2}} \\
 &= 2\pi \left[16\sqrt{2} - \frac{8 \cdot 2\sqrt{2}}{3} \right] \\
 &= 2\pi \left[16\sqrt{2} - \frac{16\sqrt{2}}{3} \right] \quad V = \frac{64\pi\sqrt{2}}{3} \text{ unit}^3
 \end{aligned}$$

Question 5

a) i) $A(3,0)$ $B(0,6)$

$$\begin{aligned}
 \text{ii) } \Delta OAB &= \frac{3 \times 6}{2} = 9 \text{ unit}^2 \\
 A_{\text{sh}} &= \int_0^3 (3-x)(2+x) - x(3-x) dx \\
 &= \int_0^3 (6+x-x^2-3x+x^2) dx \\
 &= \int_0^3 (6-2x) dx \\
 &= \left[6x - x^2 \right]_0^3 \\
 &= 18 - 9 \\
 &= 9 \text{ unit}^2 \text{ equal to } \Delta OAB
 \end{aligned}$$



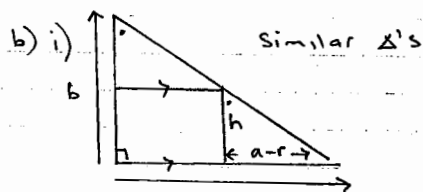
$$\begin{aligned}
 V_{\text{sh}} &= \pi \int_0^2 (4x - x^2)^2 - (x^2)^2 dx \\
 &= \pi \int_0^2 (16x^2 - 8x^3 + x^4 - x^4) dx
 \end{aligned}$$

$$\begin{aligned}
 &= \pi \int_0^2 (16x^2 - 8x^3) dx \\
 &= \pi \left[\frac{16x^3}{3} - 2x^4 \right]_0^2 \\
 &= \pi \left[\frac{128}{3} - 32 \right] \\
 &= \frac{32\pi}{3} \text{ units}^3
 \end{aligned}$$

Question 6

x	0	12	24
$f(x)$	0	28	15

$$\begin{aligned}
 \text{a) } A_s &= \frac{12}{3} [0 + 15 + 4(28)] \\
 &= 508 \text{ m}^2
 \end{aligned}$$



$$\frac{h}{b} = \frac{a-r}{a}$$

$$\therefore h = \frac{b(a-r)}{a}$$

$$\begin{aligned}
 \text{ii) } V_{\text{cylinder}} &= \pi r^2 h \\
 &= \pi r^2 b \cdot \frac{(a-r)}{a}
 \end{aligned}$$

$$\therefore V = \pi r^2 b - \frac{\pi r^3 b}{a}$$

$$\text{iii) } \frac{dV}{dr} = 2\pi r b - \frac{3\pi r^2 b}{a}$$

$$\frac{d^2V}{dr^2} = 2\pi b - \frac{6\pi r b}{a}$$

4

$$\text{st pt } \frac{dV}{dr} = 0$$

$$2\pi r b = \frac{3\pi r^2 b}{a}$$

$$2abr = 3r^2 b$$

$$2abr - 3r^2 b = 0$$

$$br(2a - 3r) = 0$$

$$\therefore r = 0 \quad 2a = 3r$$

$$\text{(no result)} \quad r = \frac{2a}{3}$$

since $r > 0$ test max/min using $\frac{d^2V}{dr^2}$

$$\text{and } r = \frac{2a}{3} \quad \frac{d^2V}{dr^2} = 2\pi b - \frac{6\pi b}{a} \cdot \frac{2a}{3}$$

$$= 2\pi b - 4\pi b$$

$$= -2\pi b$$

$$\therefore \frac{d^2V}{dr^2} < 0 \quad \text{max volume}$$

$$r = \frac{2a}{3}$$

$$\text{Max Volume} = \pi b \left(\frac{2a}{3} \right)^2 - \frac{\pi b}{a} \left(\frac{2a}{3} \right)^3$$

$$= \frac{4\pi b a^2}{9} - \frac{8\pi b a^3}{a \cdot 27}$$

$$= \frac{4\pi b a^2}{9} - \frac{8\pi b a^2}{27}$$

$$= \frac{4\pi b a^2}{27}$$

iv)

$$V_{\text{cone}} = \frac{1}{3} \pi a^2 b$$

$$\frac{4}{9} \left(\frac{1}{3} \pi a^2 b \right) = \frac{4}{27} \pi a^2 b$$

$\therefore$ Cylinder with max volume is $\frac{4}{9}$ that of cone